

Set 1 - Midterm Questions

MCQs:

1. Which command is used to list all files and directories in the current directory?

- a) ls
- b) list
- c) dir
- d) show

2. What does the 'chmod' command do?

- a) Changes file name
- b) Changes file permissions
- c) Changes directory
- d) None

3. Which symbol is used to redirect output to a file, overwriting existing content?

- a) >
- b) >>
- c) <
- d) |

4. How do you make a script executable?

- a) chmod +x script.sh
- b) exec script.sh
- c) run script.sh
- d) sh script.sh

5. What will 'echo \$HOME' print?

- a) Home directory path
- b) Username
- c) Root path
- d) Shell type

6. What is the correct syntax to read a value in bash?

- a) input var
- b) read var
- c) get var
- d) scan var

7. Which command is used to view the current working directory?

- a) path
 - b) showpath
 - c) pwd
 - d) cwd
8. What does the command 'rm -r' do?
- a) Remove read-only files
 - b) Remove recursively
 - c) Remove recently used files
 - d) None
9. What is the shebang line in a bash script?
- a) #script
 - b) #bash
 - c) #!/bin/bash
 - d) //bash
10. How do you append to a file in bash?
- a) echo 'text' >> file
 - b) echo 'text' > file
 - c) cat > file
 - d) print >> file

Written Questions:

1. Write a bash script that checks whether a number is positive, negative, or zero.
2. Write a command to create a directory named 'Midterm' and create a file inside it named 'notes.txt'.
3. Write a script to print the numbers 1 to 10 using a for loop.
4. What is the purpose of 'read' command in bash scripting? Explain with an example.

Set 2 - Midterm Questions

MCQs:

1. Which command is used to display the content of a file?
 - a) show
 - b) cat
 - c) list
 - d) read

2. What does '&&' operator do in shell?

- a) OR
- b) AND
- c) NOT
- d) NAND

3. Which command will show currently running processes?

- a) ls
- b) ps
- c) pwd
- d) whoami

4. How to run a script named test.sh in the current directory?

- a) run test.sh
- b) ./test.sh
- c) test.sh
- d) sh ./test

5. What is the default shell in most Linux systems?

- a) fish
- b) bash
- c) zsh
- d) sh

6. Which command is used to remove a file?

- a) rm
- b) del
- c) erase
- d) remove

7. Which command shows the calendar?

- a) cal
- b) calendar
- c) date
- d) time

8. Which command is used to display the first 10 lines of a file?

- a) tail
- b) head

- c) cat
- d) more

9. What does 'man' command do?

- a) Opens manual
- b) Runs a command
- c) Sends mail
- d) None

10. What does 'echo "Hello" > file.txt' do?

- a) Appends Hello
- b) Prints Hello
- c) Writes Hello to file
- d) None

Written Questions:

1. Write a script to check if a given number is even or odd.
2. Explain the difference between '>' and '>>' with example commands.
3. Write a bash script to take two numbers as input and print their sum.
4. What is the difference between 'sh script.sh' and './script.sh'?

Set 3 - Midterm Questions

MCQs:

1. What is the purpose of the 'touch' command?

- a) Delete a file
- b) Create an empty file
- c) Copy a file
- d) Rename a file

2. What does the 'grep' command do?

- a) Replace text
- b) Search for a pattern
- c) Sort a file
- d) Edit file

3. What is the output of 'expr 5 + 3'?

- a) 53
- b) 8

c) Error

d) None

4. What is the default file permission of a new file?

a) 777

b) 755

c) 644

d) 700

5. What will 'echo \$?' show?

a) PID

b) Exit status

c) User ID

d) Shell path

6. How do you comment a line in bash script?

a) -- comment

b) // comment

c) # comment

d) /* comment */

7. Which symbol is used for piping?

a) |

b) >

c) &

d) ^

8. Which command is used to rename a file?

a) rename

b) mv

c) cp

d) edit

9. What is 'cron' used for?

a) Scheduling tasks

b) Network config

c) Memory check

d) Compiling code

10. What does 'ls -l' display?

a) Long listing format

b) List in columns

c) Load average

d) Linked files only

Written Questions:

1. Write a bash script to calculate the factorial of a number.
2. Explain what happens when you type 'ls | grep txt'.
3. Write a script that checks if a file exists and prints a message accordingly.
4. Explain the difference between absolute and relative path with example.

Set 4 - Midterm Questions

MCQs:

1. What does the 'pwd' command show?
 - a) Previous directory
 - b) Present working directory
 - c) Parent directory
 - d) None
2. What does 'whoami' command do?
 - a) Tells the OS name
 - b) Shows current user
 - c) Shows hostname
 - d) Lists users
3. What does the 'cd ..' command do?
 - a) Changes to root
 - b) Moves one directory up
 - c) Deletes a directory
 - d) Opens hidden folder
4. What does '#!/bin/bash' signify in a shell script?
 - a) Path to script
 - b) Documentation
 - c) Interpreter path
 - d) Script ending
5. How to take input in a shell script?

- a) scanf
 - b) cin
 - c) read
 - d) input
6. What is the use of '\$1', '\$2' in bash?
- a) Command ID
 - b) Filename
 - c) Positional parameters
 - d) Line numbers
7. What is the output of 'echo "Hello" | wc -c'?
- a) 5
 - b) 6
 - c) 7
 - d) 8
8. What does 'tail -n 3 file.txt' show?
- a) First 3 lines
 - b) Last 3 lines
 - c) All lines
 - d) Middle 3 lines
9. Which command is used to compare two files?
- a) compare
 - b) cmp
 - c) diff
 - d) Both b and c
10. How to display system date?
- a) datetime
 - b) date
 - c) sysdate
 - d) time

Written Questions:

1. Write a script to count the number of lines in a file given as an argument.
2. Explain the difference between 'cp' and 'mv' commands.
3. Write a script to display whether a number is divisible by 5 and 11 both.

4. Explain how 'for loop' works in bash with example.